LLM12: LLM Inference — Decoding Strategies & KV-Cache

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Two-Stage Inference Pipeline

Prompt Phase (Prefill)

- Process entire prompt in parallel
- Compute-bound: high arithmetic intensity
- Latency: Time to First Token (TTFT)

Token Generation Phase (Decode)

- Generate one token at a time (sequential)
- Memory-bound: low arithmetic intensity
- Throughput: tokens per second

Autoregressive Generation

Autoregressive generation: given a prompt, generate one token at a time.

输入 $X = [x_1, x_2, \cdots, x_t]$

⇩

$$P(x_{t+1}|x_1, x_2, \cdots, x_t) = \text{Softmax}(\mathbf{h}_t \mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}}),$$

自回归 $\mathbf{h}_t \in \mathbb{R}^{d_h}$ $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d_h \times vocab}$

$$x_{t+1} \sim P(x_{t+1}|x_1, x_2, \cdots, x_t).$$

⇩

下一轮输入 $X = [x_1, \cdots, x_t, x_{t+1}]$

Why Decode is Memory-Bound

Key insight: Decode loads the **entire model weights** for just **one token**.

Model Weight Access

Stage	Weights/token
Prefill (s tokens)	$\frac{16h^2L}{s}$
Decode (1 token)	$16h^2L$

Qwen3-0.6B: weights $\approx$ 1.2 GB
At 30 tok/s: $\sim$36 GB/s bandwidth!

KV-Cache Access

Stage	KV read/step
Prefill	Write only
Decode	Read all s positions

At s=8192: $\sim$0.9 GB per decode step

Arithmetic Intensity

Arithmetic intensity = FLOPs / bytes loaded → determines compute vs. memory bottleneck.

Stage	Seq Len	FLOPs	Bytes	Ratio	Intensity
Prefill	128	92B	0.97GB	95	**compute**
Prefill	8192	13.5T	2.8GB	4779	**compute**
Decode	128	719M	0.97GB	0.74	**memory**
Decode	8192	1.6G	2.8GB	0.58	**memory**

Implication

Prefill processes many tokens in parallel → high intensity → GPU is busy.
Decode processes 1 token → low intensity → GPU is idle, waiting for data.

From Logits to Tokens (https://poloclub.github.io/transformer-explainer/)

The model outputs logits $z \in \mathbb{R}^{|\mathcal{V}|}$. How do we pick the next token?

Temperature Scaling

$$\tilde{p}_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

- $T < 1$: sharper $\rightarrow$ more deterministic
- $T > 1$: flatter $\rightarrow$ more random
- $T \rightarrow 0^+$: approaches greedy

Top-K Sampling

Keep only k highest-probability tokens, re-normalize.

Top-P (Nucleus) Sampling

Keep smallest set with cumulative prob $\geq p$.

Typical pipeline: Temperature $\rightarrow$ Top-K $\rightarrow$ Top-P $\rightarrow$ Sample

Top-K vs Top-P Comparison

Top-K (e.g., K=3)

- Sort tokens by probability
- Keep only top K tokens
- Re-normalize probabilities
- Sample from restricted set

Issue

Fixed K may be too restrictive or too loose depending on distribution shape.

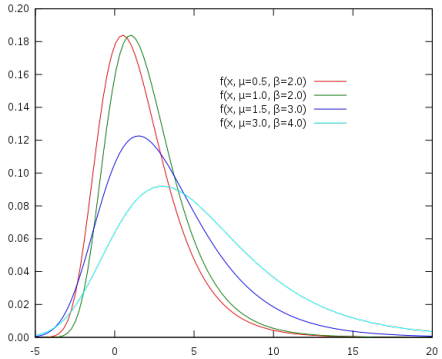
Top-P / Nucleus (e.g., P=0.9)

- Sort tokens by probability
- Accumulate until sum $\geq P$
- Dynamic candidate set size
- More adaptive than Top-K

Advantage

Automatically adjusts to distribution: few tokens for confident predictions, more for uncertain.

Temperature Effect Visualization



Temperature Control

- **Low** T: High confidence, low diversity (good for factual QA)
- **Medium** T: Balanced creativity and coherence
- **High** T: High diversity, lower quality (good for brainstorming)

Repetition Penalty Example

Models can produce repetitive text. Three penalty mechanisms:

Repetition Penalty (HuggingFace)

For each previously seen token: divide logit if positive, multiply if negative.

Presence Penalty (OpenAI)

Subtract a constant from any previously seen token's logit.

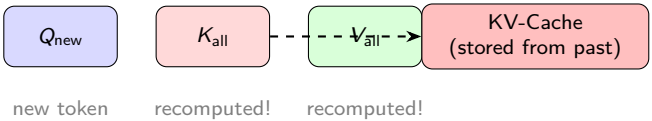
Frequency Penalty (OpenAI)

Subtract value proportional to how many times the token appeared.

Token 0 (appears 3×)	**Original**	**After penalty**
Rep penalty=1.5	3.0	2.00
Freq=0.5 + Pres=0.3	3.0	1.20

What is KV-Cache?

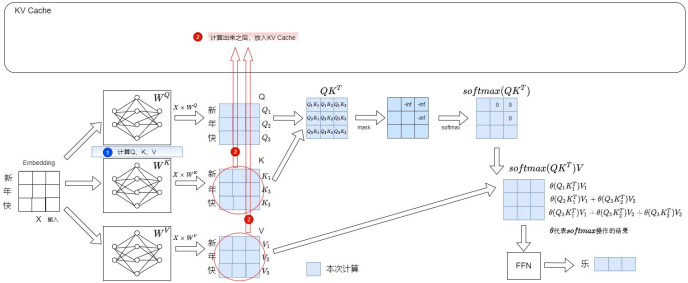
Problem: Without cache, each decode step recomputes K/V for **all** past tokens.



KV-Cache Optimization

Cache past K/V tensors. During decode, only project the **new** token's K/V.
Append to cache: $K = [K_{cache}; K_{new}]$, $V = [V_{cache}; V_{new}]$

KV-Cache Implementation

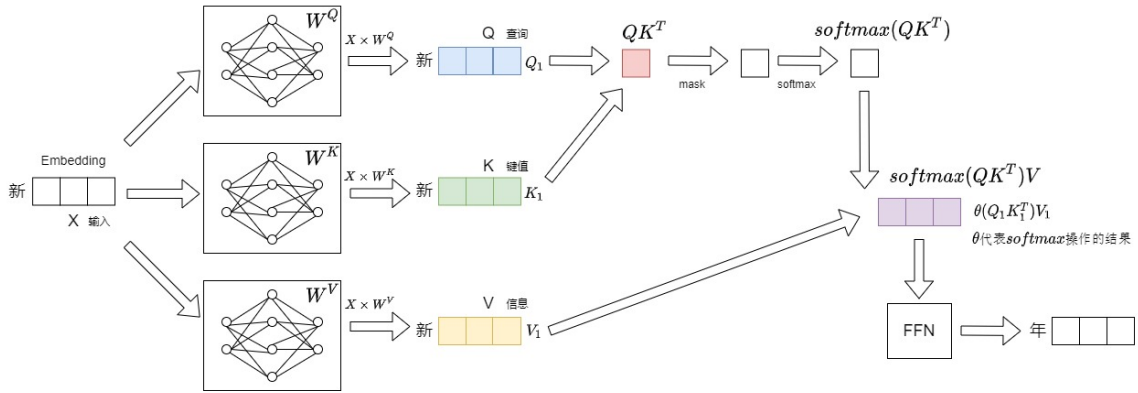


Key Steps

1. Compute Q, K, V from input embeddings
2. Store K, V in KV Cache for future reuse
3. Compute attention using cached K, V
4. Append new K, V to cache after each decode step

KV-Cache Step-by-Step Data Flow

Step 1: Input " 新" → Compute Q, K, V → Output " 年"

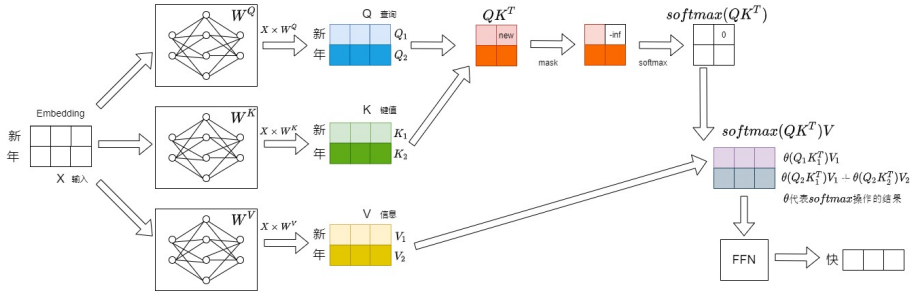


Key Observation

First token: no cache available, compute everything from scratch.

KV-Cache Step 2: Reuse Cached K/V

Step 2: Input " 新年" → Reuse cached K/V for " 新" → Output " 快"

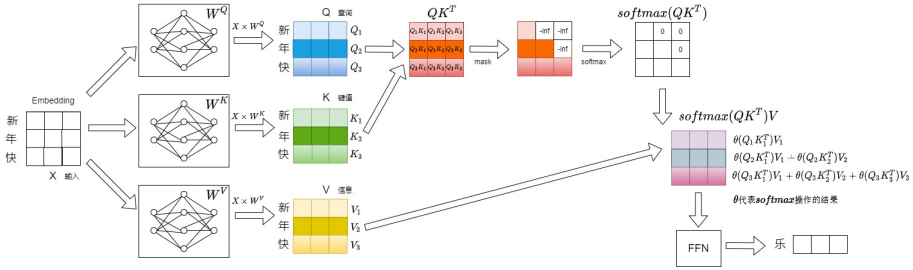


Cache Reuse

- Red circles: Repeated computation without cache
- With cache: only compute K/V for new token " 年"
- Reuse cached K/V for " 新"

KV-Cache Step 3: Cumulative Benefit

Step 3: Input " 新年快" → Reuse cached K/V for " 新年" → Output " 乐"



Growing Benefit

- Cache contains K/V for all previous tokens
- Only compute K/V for the newest token
- Complexity reduced from $O(s^2)$ to $O(s)$

KV-Cache Memory Formula

For a model with GQA (H_{kv} KV heads):

$$\text{KV memory} = 2 \times b \times L \times H_{\text{kv}} \times s \times d_{\text{head}} \times \text{dtype\_bytes}$$

Seq Length	KV Memory (FP16)	Per Layer
512	67 MB	4 MB
2,048	268 MB	17 MB
8,192	1.0 GB	67 MB
32,768	4.3 GB	268 MB
131,072	17.2 GB	1.0 GB

128K context → 17 GB KV-Cache alone!

This motivates GQA, sparse attention, and KV compression.

KV-Cache Memory Calculation Example

Example: Qwen3-0.6B with sequence length 8,192

Per-Layer KV Memory

- $B = 1$ (batch size)
- $H_{\text{kv}} = 8$ (KV heads)
- $s = 8{,}192$ (sequence length)
- $d_{\text{head}} = 128$ (head dimension)
- FP16 = 2 bytes per value

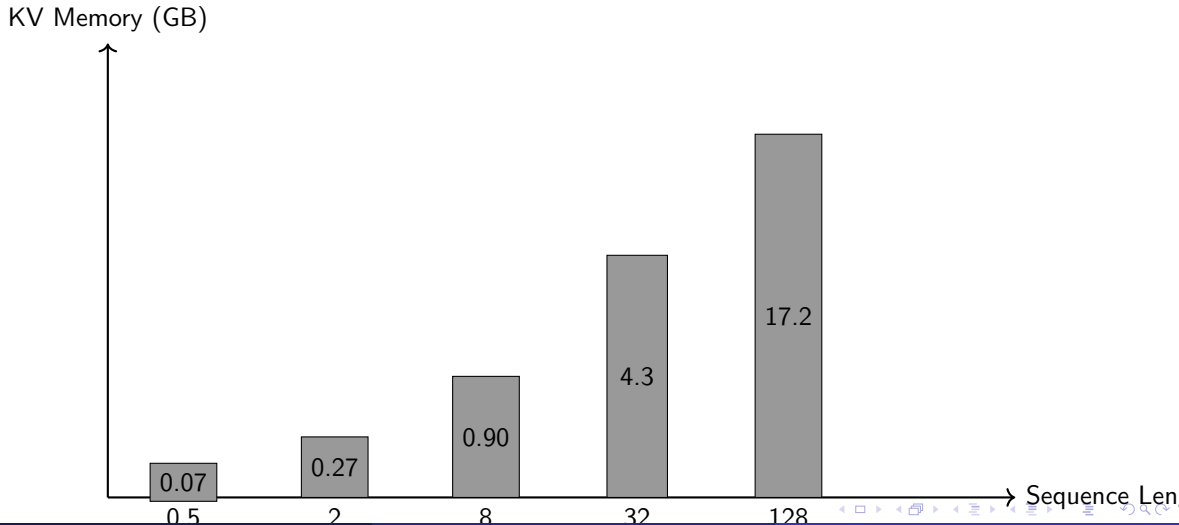
Calculation

$$\begin{aligned} \text{Per layer} &= 2 \times 1 \times 8 \times 8192 \times 128 \times 2 \\ &= 33{,}554{,}432 \text{ bytes} \approx 32 \text{ MB} \\ \text{Total (28 layers)} &= 28 \times 32 \text{ MB} \approx 896 \text{ MB} \end{aligned}$$

Key Insight

At 8K context, KV-Cache ($\sim$900 MB) is already comparable to model weights (1.2 GB)!

KV-Cache Growth Pattern



Long-Context Solutions

Technique	Reduction	Key Idea
GQA/MQA	$\times \frac{H_{kv}}{H_q}$	Fewer KV heads
FP8 KV	$\sim 2\times$	Quantize KV cache
Sliding Window	Fixed w	Attend only to last w tokens
Sparse Attention	$\sim$ 2-5$\times$	Select important KV pages
PagedAttention	No waste	Block-based memory mgmt
MLA	$\sim 10\times$	Compress KV to latent space

Batch Serving Pressure

At batch=8, seq=8192: KV-Cache (7.5 GB) **exceeds** model weights (1.2 GB)!

Qwen3-0.6B Architecture

Parameter	Value
Hidden dim	1,024
Layers	28
Q heads	16
KV heads	8
Head dim	128
Vocab size	151,936
Max context	40,960
Total params	**596M**

Measured Performance

- Prefill: 6 tokens in 467 ms
- Decode: 30 tokens, avg 24 ms/token $\rightarrow$ 42 tok/s
- KV-Cache: 0.69 MB (28 layers, 6 tokens)

Real-World Serving Systems

System	Key Features	KV Optimization
vLLM	PagedAttention, continuous batching	Block-based KV mgmt
TensorRT-LLM	Graph optimization, quantization	FP8 KV, page attention
SGLang	RadixAttention, structured output	KV cache reuse
TGI	Multi-device, streaming	Efficient batching

vLLM (Most Popular)

- PagedAttention eliminates memory fragmentation
- Continuous batching: process requests as they arrive
- 2-24$\times$ throughput improvement over naive serving

Performance Benchmarking

Key Metrics for LLM inference:

Latency Metrics

- **TTFT** (Time to First Token): How long until response starts
- **TPOT** (Time Per Output Token): Average time per token
- **Latency** = TTFT + (tokens × TPOT)

Throughput Metrics

- **Tokens/sec**: Total tokens generated per second
- **Requests/sec**: Number of completed requests
- **Concurrency**: How many requests handled simultaneously

Trade-off

Higher batch size → better throughput, but higher latency per request.
Optimal batch size depends on SLA requirements.

Summary

1. **Two-stage pipeline**: Prefill (compute-bound) vs Decode (memory-bound)
2. **Arithmetic intensity**: Decode loads entire model for 1 token $\rightarrow$ very inefficient
3. **Decoding strategies**: Temperature, Top-K, Top-P control diversity vs determinism
4. **KV-Cache**: Essential optimization — cache past K/V to avoid $O(s^2)$ recomputation
5. **GQA**: Halves KV-Cache size with minimal quality loss
6. **Long-context bottleneck**: KV-Cache dominates memory at 32K+ tokens
7. **MTP**: Predict multiple tokens per pass — no separate draft model needed
8. **Advanced techniques**: PagedAttention, Flash Attention, quantization enable efficient serving

Experiment Further

- Train MTP heads and measure real speedup
- Try FP8 KV-Cache quantization
- Compare streaming latency (TTFT vs TPS)
- Benchmark vLLM vs naive serving

Further Reading

- **FlashAttention**: Fast and Memory-Efficient Exact Attention (Dao et al., 2022)
- **vLLM**: Easy LLM Inference Serving (Kwon et al., 2023)
- **DeepSeek-V3**: Multi-Token Prediction (DeepSeek, 2024)
- **GQA**: Training Generalized Multi-Query Models (Ainslie et al., 2023)
- **PagedAttention**: vLLM's memory management technique

Resources

- GitHub: github.com/vllm-project/vllm
- HuggingFace: huggingface.co/docs/transformers
- AMD ROCm: rocm.software

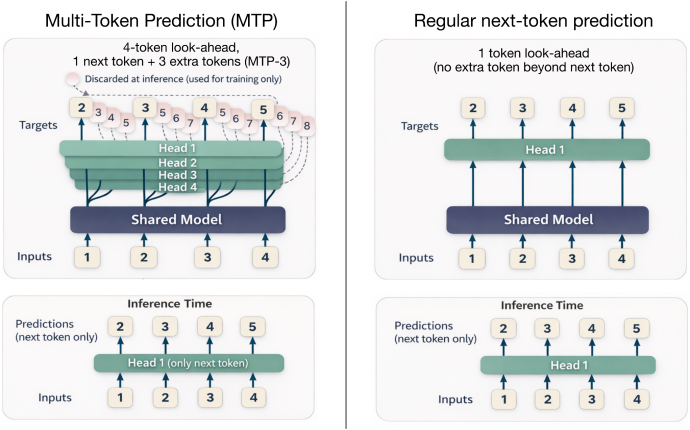
Standard vs Multi-Token Prediction

	Standard	**MTP**
Training target	$t+1$ only	$t+1, t+2, \ldots, t+n$
Prediction heads	1 shared	1 trunk + n heads
Tokens per pass	1	n
Draft model needed	Yes	No
Extra memory	Full model	Small heads only

Key Insight

The model's hidden state encodes enough information to forecast multiple future tokens. MTP is a **learnable skill** — no separate draft model needed.

MTP Architecture



- Each head: Linear(hidden_dim, vocab_size) — same as `lm_head`
- Qwen3-0.6B + 4 heads: 596M + 467M params
- In production (DeepSeek-V3): $\sim$1.8$\times$ speedup

MTP Inference Flow

One forward pass $\rightarrow$ n tokens!

1. Run trunk $\rightarrow$ get hidden state h_t
2. Run all n heads in parallel $\rightarrow$ predictions for $t+1, t+2, \ldots, t+n$
3. Accept tokens greedily (or verify with confidence threshold)
4. Continue from position $t+n$ (skip $n-1$ decode steps)

Comparison with Speculative Decoding

- Speculative: needs a separate draft model (extra memory + complexity)
- MTP: model is its own draft (only adds small projection heads)
- Both reduce the number of autoregressive decode steps

Limitation

MTP heads must be trained on multi-token targets.
Randomly initialized heads produce garbage predictions.

MTP Training Objective

Standard Training

- Single prediction head
- Target: next token $t+1$
- Loss: cross-entropy on $t+1$

$$\mathcal{L} = -\log P(x_{t+1}|x_{\leq t})$$

MTP Training

- Multiple prediction heads
- Targets: $t+1, t+2, \ldots, t+n$
- Combined loss across all heads

$$\mathcal{L} = \sum_{i=1}^{n} w_i \cdot \mathcal{L}_i$$

$$\mathcal{L}_i = -\log P(x_{t+i}|x_{\leq t})$$

Key Difference

MTP heads learn to predict **multiple future tokens** from the same hidden state.
This requires the model to encode richer predictive information.

MTP vs Speculative Decoding

Aspect	MTP	Speculative
Draft model	Built-in	Separate small model
Training cost	Higher (multi-head)	Train draft separately
Inference memory	+small heads	+full draft model
Speedup	$\sim$1.5-2$\times$	$\sim$2-3$\times$
Quality	Same as base	May differ (draft bias)
Complexity	Lower	Higher (verify logic)

When to Use Which?

- **MTP**: When you can train/finetune the model, memory-constrained
- **Speculative**: When using frozen models, need maximum speedup